

Question:

Carla is downloading a 200 GB file. Normally she can download 2 GB/minute, but 40% of the way through the download, Windows forces a restart to install updates, which takes 20 minutes. Then Carla has to restart the download from the beginning. How long does it take to download the file?

Reasoning + Answer:

<think> Carla is downloading a 200 GB file. So, normal downloading speed is 2 GB / minute. This means, to download 200 GB will take = $200 / 2 = 100$ minutes. Now, a restart is forced 40% of the way through the download. So, amount of data downloaded before restart is 40% of 200 GB which is $(40 / 100) * 200 = 80$ GB. So, time taken to download 80 GB is $80 / 2 = 40$ minutes. Adding this to the restart time of 20 minutes, total time so far is $40 + 20 = 60$ minutes. Now, she has to restart the download from the beginning. So, entire process will take original time which is 100 minutes. Therefore, total time taken to download the file is 100 minutes</think><answer>100</answer>

